Chapter 7 Problem Set

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1. We know that if $ab \equiv ac \pmod{m}$, $b \equiv c \pmod{\frac{m}{\gcd(a,m)}}$. In our case, since $6x \equiv 4 \pmod{28}$, we know that $3x \equiv 2 \pmod{14}$. And because $2 \equiv 30 \pmod{14}$, we can see that $3x \equiv 30 \pmod{14}$, so $x \equiv 10 \pmod{14}$.

2. Essentially, we want to find an integer x such that $ax \equiv 1 \pmod{m}$. We are given that $\gcd(a, m) = 1$, so by Bezout's Lemma, there exist integers x and y such that $ax + by = 1$. We can use the Euclidean algorithm on a and m to find x and y. The value of x that we found is equivalent to a modulo m.

To compute $\frac{1}{14} \pmod{107}$, we are looking for an integer x such that $14x \equiv 1 \pmod{107}$. Now, we can run the Euclidean Algorithm on a and m. We have:

$$9 = 107 - 14 \cdot 7$$

$$5 = 14 - 9 \cdot 1$$

$$4 = 9 - 5 \cdot 1$$

$$1 = 5 - 4 \cdot 1.$$

Substituting backward multiple times, we get

$$1 = 5{-}4{\cdot}1 = 5{-}(9{-}5{\cdot}1) = 2{\cdot}5{-}9 = 2(14{-}9{\cdot}1){-}9 = 2{\cdot}14{-}3{\cdot}9 = 2{\cdot}14{-}3(107{-}14{\cdot}7) = 23{\cdot}14{-}3{\cdot}107.$$

We see that $23 \cdot 14 - 3 \cdot 107 = 1$, so the value of x is 23.

3. We are trying to prove that the only values of x such that $x \equiv \frac{1}{x} \pmod{p}$ are ± 1. That equivalence is the same as saying that $x^2 \equiv 1 \pmod{p}$. That means $x^2 - 1 \equiv (x-1)(x+1) \pmod{p}$. Since p is prime, $x-1$ and $x+1$ can only be 0 or multiples of p. That means either $x - 1 \equiv 0 \pmod{p}$ or $x + 1 \equiv 0 \pmod{p}$, so $x \equiv \pm 1 \pmod{p}$.

4. Let's say that $a = ga'$ and $b = gb'$ where $\gcd(a', b') = 1$. Since $\gcd(a', b') \operatorname{lcm}(a', b') = a'b'$, we have that $\operatorname{lcm}(a', b') = \frac{ab}{g^2}$, and also, since a' and b' are relatively prime, we know that $\operatorname{lcm}(a', b') = a'b'$. Therefore, $ab = g^2 a'b'$.

Now, because the totient function is multiplicative and each of g^2, a', and b' are relatively prime to each other, we know that $\phi(ab) = \phi(g^2a'b') = \phi(g^2)\phi(a')\phi(b')$. Earlier, we stated that $a = ga'$ and $b = gb'$, which means $\phi(a) = \phi(g)\phi(a')$ and $\phi(b) = \phi(g)\phi(b')$. Isolating $\phi(a')$ and $\phi(b')$ and substituting into our earlier equation, we have $\phi(ab) = \frac{\phi(g^2)\phi(a)\phi(b)}{(\phi(g))^2}$.

To proceed, I would have to prove that $\phi(g^2) = g\phi(g)$, but I couldn't figure out how to do so.